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Investigating subjective and objective sleep in functional neurological disorder using self-reports and actigraphy – A cross-sectional study

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ABSTRACT

Background: Functional neurological disorder (FND) is a frequent and disabling neuropsychiatric condition. Patients frequently report comorbid sleep disturbances; however, the role and aetiology of sleep impairments in FND remain incompletely understood.

Methods: We assessed sleep quality in a cohort of mixed FND patients and healthy controls (HC). All participants completed the Pittsburgh Sleep Quality Index (PSQI; subjective), whereas patients additionally underwent a 2-week actigraphy (objective). First, we compared subjective sleep quality between groups and assessed the role of comorbid psychiatric symptoms. Secondly, in patients only, we evaluated subjective and objective sleep quality according to clinically meaningful thresholds and examined the inter-assessment concordance (subjective vs. objective measures).

Results: 26 mixed FND patients (73% female, mean age 38.4y) and 26 age- and sex-matched HC (73% female, mean age 36.5y) were included. We observed significantly impaired subjective sleep quality (sleep efficiency, sleep latency, sleep disturbances, daytime dysfunction, overall sleep quality) in FND compared to HC, whereas symptoms of depression and anxiety appeared as contributing factors. Sleep quality was classified as bad to chronically disturbed in 77% of patients compared to 31% of HC. Across modalities, sleep efficiency qualified as clinically relevant in most patients (62% subjective, 73% objective), whereas we observed a significant inter-assessment discrepancy in sleep latency ($p = 0.001$) in FND.

Conclusion: This study depicts sleep impairments in FND patients compared to a healthy control sample and according to clinically meaningful thresholds, in addition to a disparity between subjective and objective perception. These findings suggest sleep as a critical comorbidity in FND, which warrants greater attention.

1. Introduction

The term functional neurological disorder (FND) refers to a neuropsychiatric disorder in which patients present with neurological symptoms compatible with brain network dysfunctions [1]. FND is one of the most common reasons for a referral to neurology [1] and severely affects patients' well-being and social functioning [2–4]. FND presents with various phenotypes and multi-faceted co-morbidities, including sleep disturbances [5]. A recent meta-analysis reported sleep impairment in 58% of patients [6], while individual studies found rates of 46.7% [5],

Abbreviations: AASM, American Academy of Sleep Medicine; BDI, Beck Depression Inventory; FND, Functional Neurological Disorder; FS, Functional Seizures; HC, healthy controls; PNES, Psychogenic Non-Epileptic Seizures; PSQI, Pittsburgh Sleep Quality Index; REM, Rapid-eye-movement; SE, sleep efficiency; S-FMDRS, Simplified Functional Movement Disorders Rating Scale; SL, sleep latency; SSS, subjective symptom severity; STAI, State and Trait Anxiety Inventory; TST, total sleep time.

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62% [7] and up to 89% [8]. Furthermore, sleep disturbances have been found to be associated with functional impairment [8] and quality of life in FND patients [9]. Besides the influence of sleep on general mental health [10–12], in FND, poor sleep may further affect overall well-being and functional symptoms through altered sensory processing and interoception [5], elevated dissociative states or altered sense of agency [6,8], increased fatigue or a perpetuating bidirectional worsening of psychiatric comorbidities like depression and anxiety [13].

Despite this assumed impact of sleep and repeated reports of frequent sleep disturbances in FND, the role and origin of sleep impairments remain unclear in the current literature, arguably linked to four major challenges: Firstly, most previous studies were limited to small samples and predominantly included patients with functional seizures (FS; previously psychogenic non-epileptic seizures, PNES). However, FND encompasses several phenotypes, and only the inclusion of mixed samples will extend the understanding across symptom presentations. Secondly, patients' sleep quality has been compared to diverse control groups, limiting conclusions to relative differences and does not clarify whether the observed alterations reflect clinically meaningful impairments. Thirdly, given frequent comorbid depression and anxiety, a debate on whether sleep disturbances are emerging from psychiatric comorbidities rather than FND itself is ongoing, but studies often refrain from testing this possible association. Lastly, whether patients' reports of impaired sleep reflect objectively altered sleep patterns or rather patients' aberrant perception of their own sleep quality in line with impaired interoceptive skills remains debated. Comparing subjective and objective reports is required to address this question, yet many studies focus on only one modality.

Essentially, recent work highlighted the scarcity of literature and the striking heterogeneity in methods [6,14], calling for further research to shed light on the role of sleep in FND. Addressing the above-mentioned challenges, we set out to investigate sleep patterns in a mixed sample of FND patients using subjective and objective measures. We aimed to (1a) compare subjective measures of sleep quality in FND patients to HC, and (1b) exploratorily testing the association with psychiatric symptoms, (2) evaluate FND patients' subjective and objective sleep parameters in regard to literature-based, clinically meaningful thresholds for distinguishing healthy from impaired sleep, and lastly (3) compare the objective and subjective measure within the FND cohort to elaborate on a potential imprecise perception of one's own sleep. While we expected to observe poor sleep quality when compared to HC (aim 1a), as well as in relation to literature-based thresholds delineating healthy sleep (aim 2), the contributions of psychiatric symptoms (aim 1b) and inter-assessment concordance (aim 3) were analysed with an exploratory approach due to the lack of definitive evidence in previous literature.

2. Methods

The study was part of a cross-sectional study investigating biomarkers for FND [15–22], approved by the Ethics Committee of Canton Bern, Switzerland (2017–00997, DRKS00012992), conducted in accordance with the Declaration of Helsinki.

2.1. Participants

Patients with a diagnosis of FND based on positive signs [23] and according to DSM-5 [2] were recruited from the Psychosomatics Unit, Department of Neurology, University Hospital Inselspital Bern, Switzerland. Inclusion criteria were age above 16 years and capability of judgment. Exclusion criteria were severe comorbid psychiatric disorders with acute risk (e.g., acute suicidality), inability to follow the study procedures and contraindication to MRI (due to overarching study). HC were recruited with the same inclusion and exclusion criteria applied. All participants provided written informed consent and answered a questionnaire on sleep quality. Due to specific interests in sleep patterns in FND, patients were additionally offered actigraphy during a 14-day period. For this study, a matched case-control design was employed, where FND patients with completed actigraphy data were individually matched to HC based on sex and age.

2.2. Materials

2.2.1. Actigraphy

Objective sleep data of FND patients was assessed during 14 days using wrist-worn accelerometers (MotionWatch 8, CamNtech Ltd., Cambridgeshire, UK) and pre-processed according to the user guide (MotionWare 1.2.50) [24], specifying an epoch length of 60 s and uniaxial mode. Actigraphy events marking (e.g., lights off, getting up, naps, wear-off periods) were based on (1) movement and (2) light detection, and (3) the assistance of patient-recorded sleep diaries. Nights with missing data of >3 h were excluded, non-wear periods of >10 min and < 3 h during the day were substituted with the average of that day and naps were marked. The parameters total sleep time (TST; total duration of time effectively slept), sleep latency (SL; time between switching off the lights and falling asleep), and sleep efficiency (SE; percentage of total sleep duration relative to time spent in bed) were included for analysis.

2.2.2. Questionnaires

Subjective sleep was assessed with the Pittsburgh Sleep Quality Index (PSQI; 19 self-rated items) [25], and aggregated according to the scoring manual, to seven sleep components (subjective sleep quality, SL, sleep duration, SE, sleep disturbances, use of sleep medication, daytime dysfunction; possible range 0–3) and a total score (possible range 0–21). The questionnaire results are reported in the recommended scoring (components: 0 = healthy, 1–3 = impaired; total score: 0–5 = healthy, 6–10 = impaired, 11–21 = chronic sleep problems). The components TST, SL, and SE were additionally analysed in their unscored numeric form (i.e., hours, minutes, and percentage, respectively) to allow comparison with actigraphy. Mood was assessed with the Beck Depression Inventory (BDI) [26] and the State and Trait Anxiety Inventory (STAI) [27]. FND symptom severity was assessed by the Simplified Functional Movement Disorders Rating Scale (S-FMDRS) [28] and self-reported subjective symptom severity (SSS; visual analogue scale, 0–100) in patients only.

2.3. Data analysis

Comparing FND patients to HC regarding demographic and clinical characteristics, a Chi-Square test was used to test for gender distribution and *t*-test or Wilcoxon-rank-sum-test to compare age, BDI and STAI-trait scores, as indicated after testing for normal distribution (using Shapiro-Wilk test).

First, subjective sleep quality measures (PSQI) were compared between FND patients and HC. Normality of component scores and numeric variables (sleep time [h], SL [min], SE [%]) was tested (Shapiro-Wilk), followed by a *t*-test or Wilcoxon-rank-sum-test as indicated and FDR-correction to account for multiple comparisons, along with appropriate effect size (Cohen's *d* or *r*, respectively [29]). Exploratorily, we tested whether depression (BDI) and anxiety (STAI-trait) symptoms explained sleep quality beyond group: for **component scores**, cumulative link models (ordinal logistic regression with logit link for categorial levels; *ordinal* package) were fitted, comparing null (group) and full (group, depression, anxiety) models and testing proportional odds assumption in full models (Brant test; *brant* package). For **numeric variables**, linear models (*lm* package) were built analogously, testing for linearity, normality, multicollinearity and homoscedasticity. Model fit was assessed with Akaike information criterion (AIC, $\Delta > -2$ indicating better fit) and model comparisons with ANOVA (*LR* X^2 for ordinal, F-statistics for linear), while reporting the prediction values of the variables group, depression and anxiety in the full model. When assumptions were violated, model comparisons were reported for

completeness (to be interpreted with caution), and prediction values were reported using HC3 heteroskedasticity-consistent standard errors (*sandwich*, *lmtest*). Due to the exploratory nature, no correction for multiple comparisons was applied. Additional exploratory analyses on potential interaction effects are reported in Supplementary Material A.

Second, the subjective and objective sleep quality (actigraphy) was evaluated according to literature-based, clinically meaningful thresholds for distinguishing healthy from impaired sleep: the percentage of subjects having a healthy sleep behaviour was determined by the cut-offs for TST >7 h [30], SE >85%, and SL <30 min [31].

Third, the degree of concordance between subjective and objective measures in FND patients was determined using a paired *t*-test or Wilcoxon-signed-rank-test after testing for normal distribution of the data (Shapiro-Wilk test). All statistics were performed in R (version 4.5.0) [32] with a significance level set at $\alpha = 0.05$.

3. Results

3.1. Sample

For this study, complete data from 26 FND patients was obtained and compared to 26 age- and sex-matched HC (see Fig. 1). Demographic and clinical details are reported in Table 1.

3.2. Subjective sleep in FND vs. HC

Due to non-normality in all variables, Wilcoxon-rank-sum tests (FDR-corrected for multiple comparisons) were applied for comparing PSQI self-reports between groups. FND patients revealed a significantly reduced subjective sleep quality ($W = 452$, $p = 0.037$, $r = 0.306$), SE ($W = 482$, $p = 0.008$, $r = 0.415$), as well as increased SL ($W = 467$, $p = 0.024$, $r = 0.339$), sleep disturbances ($W = 465$, $p = 0.009$, $r = 0.397$) and daytime dysfunction ($W = 515$, $p = 0.003$, $r = 0.476$) compared to HC, with no significant group differences in sleep duration ($W = 321$, $p = 0.730$, $r = 0.049$) or use of sleep medication ($W = 389$, $p = 0.228$, $r = 0.180$; Fig. 2).

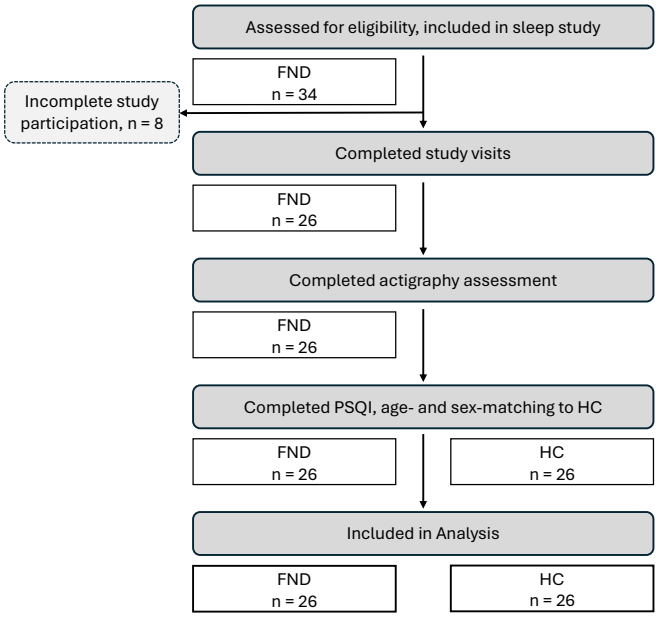

Fig. 1. Inclusion flow-chart, illustrating eligibility assessment, inclusion and exclusion, and age- and sex matching of HC to FND patients with complete data in a case-control design. Incomplete study participation due to lack of time/temporal unavailability ($n = 2$), aggravation of symptoms (n = 2), COVID-19 ($n = 1$), no reason provided ($n = 3$). Data acquisition occurred between 2020 and 2021.

Table 1
Participants' demographic and clinical characteristics.

Participant characteristics FND vs. HC			
Variable	Group		p-value[2]
	FND, N = 26[1]	HC, N = 26[1]	
Demographics			
Sex, *n (%)*			*>0.9*
Male	7 (27%)	7 (27%)	
Female	19 (73%)	19 (73%)	
Age, *mean (SD)*	38.4 (11.5)	36.5 (10.5)	*0.5*
Clinics & Mood			
Symptom Type[3]	14 Weakness 9 Gait Disorder 1 Tremor 6 Myoclonus, 5 Functional seizures 2 Sensory (pure) 4 Speech Disorder 1 Dystonia 1 Functional deafness		
Subjective Symptom Severity (SSS; 1–100), *mean (SD)*	49 (30)		
Objective Symptom Severity (S-FMDRS), *median (IQR)*	6.5 (0,12)		
Beck Depression Inventory (BDI), *mean (SD)*	11.6 (7.0)	5.2 (8.4)	*<0.001***
State-Trait-Anxiety Inventory (STAI; trait), *mean (SD)*	42.5 (9.5)	34.2 (7.9)	*0.002***

S-FMDRS: Simplified Functional Movement Disorders Rating Scale.
[1] n (%): sex; Mean (SD): age, SSS, BDI, STAI-trait; Median (IQR): S-FMDRS.
[2] X^2-test: sex; two-sample t-test: age; Wilcoxon-rank-sum-test: BDI, STAI-trait, *: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$.
[3] Patients could present with >1 symptom type.

Furthermore, PSQI total score analysis revealed significantly impaired overall sleep quality in FND (Table 2), with 38.5% of the FND patients being categorised as having chronic sleep, 38.5% qualified as poor sleepers and only 23% as healthy sleepers. In contrast, the control sample consisted of 69% healthy sleepers and equally 15.5% as having bad or chronically disturbed sleep. Analysis of unscored variables revealed increased SL [min], increased time in bed [h] and significantly reduced SE [%] in FND compared to HC, but no significant group differences in sleep duration [h] (Table 2).

For all components except for the use of medication, model comparisons (ΔAIC and ANOVA) hint towards model superiority for the model including depression and anxiety scores (Table 3). In linear regressions, model fit comparisons (ΔAIC) reveal improved fit for all models when including comorbid psychiatric symptoms compared to group (FND vs. HC) alone, except for SL (Table 3). Across all variables, neither depression nor anxiety scores were consistently associated with the sleep parameter in the full model. For further exploratory analysis investigating potential interaction effects, see Supplementary Material A.

3.3. Clinically meaningful threshold and inter-assessment discordance in FND

Actigraphy data from FND patients from a mean of 12.7 nights (*Mdn* = 13 nights, *range* = 8–14 nights; see further details in Supplementary Material B) were included in the analysis. Jointly evaluating objective and subjective sleep assessments in FND patients, TST was found to exceed the thresholds for healthy sleep for 31% in subjective and 58% in objective evaluation. SE was impaired for 62% in subjective and 73% in

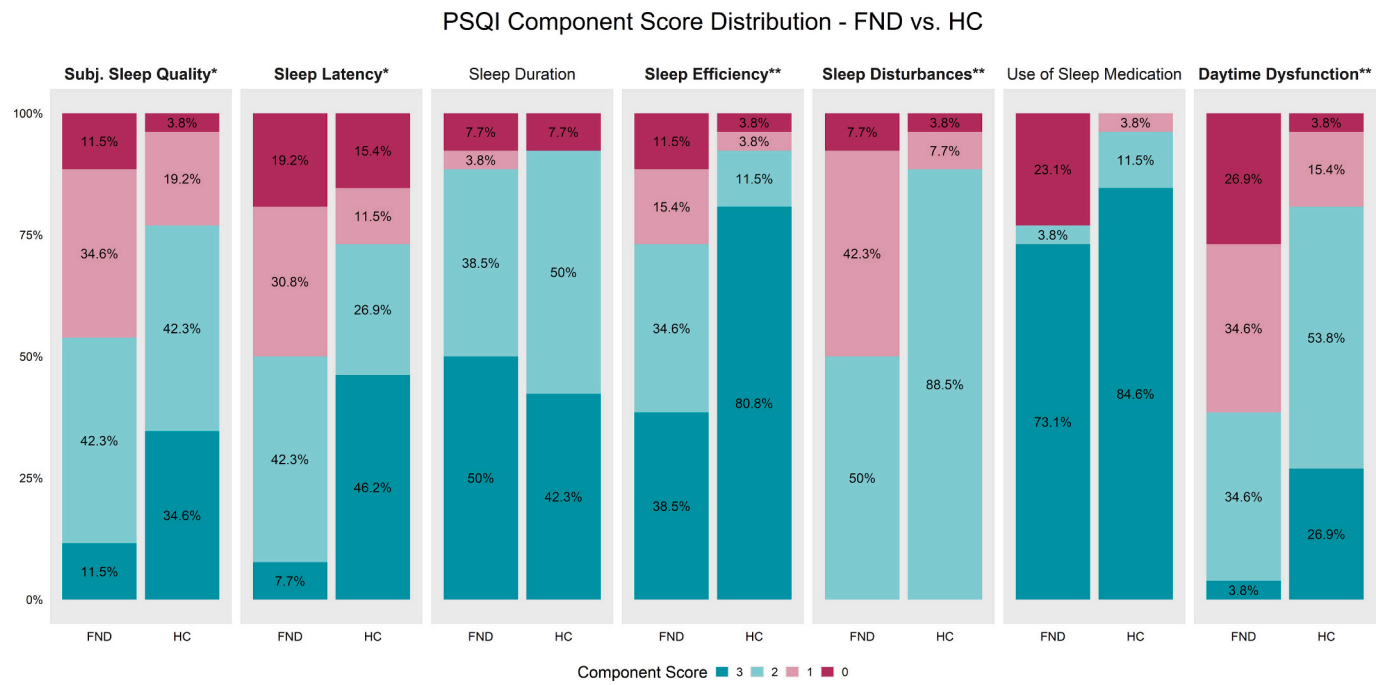

Fig. 2. Subjective sleep (PSQI) in FND patients and HC, visualised as distribution (in percentage [%]) of PSQI component scores [range: 0–3]. Higher values indicating worse sleep quality (Buysse et al., 1989). *: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$ (FDR-corrected). FND: Functional neurological disorder; HC: healthy controls.

Table 2
Subjective sleep (PSQI) compared between FND patients and healthy controls (HC) – continuous variables.

Variable	Group		Statistics		
	FND, N = 26[2]	HC, N = 26[2]	W	p-value[3]	r[4]
Component Scores					
Total Score[1]	8.5 [6.0, 12.0]	4.0 [3.0, 7.8]	522	*0.003***	0.468
Healthy sleepers, n (%)	6 (23%)	18 (69%)			
Poor sleepers, n (%)	10 (38.5%)	4 (15.5%)			
Chronic sleep disturbances, n (%)	10 (38.5%)	4 (15.5%)			
Unscored Values					
Sleep Latency [min]	30 [15, 38]	10 [5, 20]	472	*0.028**	0.342
Sleep Duration [h]	7.12 [6.12, 8.00]	7.00 [6.77, 7.50]	368	*0.592*	0.076
Time in Bed [h]	8.50 [7.81, 9.88]	8.00 [7.31, 8.50]	455	*0.043**	0.299
Sleep Efficiency [%]	79 [73, 93]	93 [88, 94]	200	*0.028**	0.350

[1] Total Score interpretation: 0–5 = healthy sleepers, 6–10 = poor sleepers; >10 = chronic sleep disturbances (Buysse et al., 1989).
[2] Median (IQR).
[3] Wilcoxon-rank-sum-test, *: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$; FDR-corrected for multiple comparisons for all component scores and unscored values, respectively.
[4] Effect size r, 0.10 - < 0.3: small effect; 0.30 - < 0.5: moderate effect and ≥ 0.5: large effect.

objective evaluation, whereas subjective SL qualified for sleep quality impairment in 27% of patients' subjective reports and 4% of objective assessments (Table 4). Lastly, FND patients demonstrated a significant inter-assessment divergence by a perceptual overestimation of SL, but not TST or SE (Table 4).

4. Discussion

In this study, we observed a significantly reduced subjective sleep quality in FND patients compared to HC, with mood symptoms appearing as contributing factors. Additionally, clinically meaningful sleep disturbances were detected in FND in both objective (TST, SL) and subjective assessments (TST, SL, SE). Lastly, we identified an inter-assessment discordance in SL in FND patients.

4.1. Sleep in FND

4.1.1. Subjective

In subjective sleep measures (PSQI), our study revealed significantly worse sleep quality in FND patients according to a total score as well as in 5 out of 7 specific domains when comparing to an age- and sex-matched control cohort, including the components subjective sleep quality, SL, SE, sleep disturbances and daytime dysfunction. These results confirm prior reports of impaired sleep in FND: poor sleep quality was observed in a large study comparing FS versus epilepsy in self-reports (BDI-II, item 16) [9]. The same group consolidated this observation in a smaller cohort based on an extended questionnaire battery [33], where 69% of the FS cohort reported very bad to fairly bad sleep quality. Similarly, in an online survey including 205 FND patients, Graham and Kyle [8] observed poor sleep quality (89% in clinical range indicating insomnia). Lastly, significantly reduced sleep quality based on PSQI was also reported in FS patients compared to HC [34].

Likewise, our findings regarding SL and SE are corroborated by earlier reports of prolonged SL in FS compared to epilepsy patients [33], while also SE was previously found to be reduced (mean = 64.4%) in FND [8].

Although not reaching statistical significance in group comparison, the use of sleep medication is remarkable in the patient cohort, with up to 23% reporting the use of three or more times a week. Similarly, in previous studies, FS patients reported significantly higher use of sleep medication (44%) compared to epilepsy patients (0%) [33], while in another cohort, 34.6% reported sleep medication at least once a week [8].

Table 3
Exploratory model statistics of ordinal or linear regression models accounting for comorbid psychiatric symptoms.

Parameter	Model fit [AIC][1]			Statistics[2]		Coefficients association (*p*-value)		
	Group	Group + depression (BDI) + anxiety (STAI)	ΔAIC	LR X^2; F (df = 2)	p-value	Group	BDI	STAI
Ordinal Regression[3]								
Subjective Sleep Quality	133.17	123.80	9.4	13.376	*0.001***	*0.502*	*0.015**	*0.718*
Sleep Latency	142.23	135.91	6.3	10.32	*0.006***	*0.210*	*0.106*	*0.298*
Sleep Duration	110.93	107.03	3.9	7.898	*0.019**	*0.088*	*0.229*	*0.271*
Sleep Efficiency	109.51	100.79	8.7	12.715	*0.002***	*0.082*	*0.156*	*0.002***
Sleep Disturbances	76.677	59.754	16.9	20.924	*<0.001****	*0.340*	*0.005***	*0.446*
Use of Sleep Medication	80.093	80.038	0.1	4.055	*0.132*	*0.305*	*0.134*	*0.061*
Daytime Dysfunction	128.45	122.89	5.6	9.553	*0.008***	*0.025**	*0.265*	*0.109*
Linear Regression[4]								
Total Score	291.96	276.89	15.1	10.632	*<0.001****	*0.138*	*0.139*	*0.031**
Sleep Latency [min]	506.27	508.00	−1.7	1.069	*0.352*	*0.452*	*0.685*	*0.536*
Sleep Duration [h]	187.95	178.05	9.9	7.354	*0.002***	*0.069*	*0.045**	*0.393*
Time in Bed [h]	201.74	195.95	5.8	4.972	*0.011**	*0.006***	*0.033**	*0.261*
Sleep Efficiency [%]	422.45	416.99	5.5	4.788	0.013*	*0.170*	*0.228*	*0.013**

BDI: Beck Depression Inventory; STAI: State and Trait Anxiety Inventory, trait *: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$.
[1] Akaike information criterion (AIC) represents model fit. Lower values indicate better model fit with consideration of model complexity; a difference > 2 indicates meaningful improvement.
[2] likelihood ratio (LR) test statistics and X^2 for ordinal regression, F for linear regression.
[3] The proportional-odds assumption (Brant omnibus test) was met for all models in the ordinal regression.
[4] All models violated the normality assumptions, while the model for Time in Bed further violated multicollinearity. While AIC-based model comparison remains valid, F-tests are reported for completeness but should be interpreted with caution, *p*-values for coefficients are based on HC3 heteroskedasticity-consistent standard errors.

Table 4
Proportion of FND patients exceeding literature-based thresholds delineating healthy sleep, descriptive statistics of subjective and objective assessments and the statistical significance of inter-assessment difference.

Parameter	Prop. patients outside of healthy range[1]		Median (IQR)		Statistics[2]		
	Subj. [%]	Obj. [%]	Subj.	Obj.	W	p-value	r[3]
Total Sleep Time [h]	31	58	7.13 [6.12, 8.00]	6.74 [6.46, 7.65]	192	*0.689*	0.082
Sleep Latency [min]	27	4	30.00 [15.00, 37.50]	15.69 [8.44, 18.04]	299	*0.001***	0.615
Sleep Efficiency [%]	62	73	79.27 [73.30, 93.08]	80.10 [77.43, 85.03]	184	*0.842*	0.042

[1] The percentage of subjects having a non-healthy/impaired sleep was determined by the literature-based threshold for TST >7 h (Watson et al., 2015), SE >85%, and SL <30 min (Schutte-Rodin et al., 2008).
[2] Wilcoxon-signed-rank-test (at least one of the two variables was non-normally distributed), *: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$.
[3] Effect size r, 0.10 - < 0.3: small effect; 0.30 - < 0.5: moderate effect and ≥ 0.5: large effect.

While our sample presented with no significant group difference regarding TST, nearly one-third of the patients exhibited TST below recommended thresholds for healthy sleep [30], which matches previous subjective reports of short TST (mean = 6.05 h) [8] in FND. Furthermore, 27% of patients in our study exceeded literature-based cut-off values for clinically meaningful sleep impairments [31] regarding SL and 60% regarding SE, which further extends the above-discussed evidence of altered sleep patterns in FND based on absolute criteria rather than relative group comparisons.

Taken together, our results match previous findings of impaired subjective sleep in multiple domains, reinforcing the notion of poor subjective sleep in FND and further adding new evidence for increased sleep disturbances and daytime dysfunction that await replication.

4.1.2. Objective

Aiming at obtaining an objective and simultaneously ecologically valid measure of sleep, we employed actigraphy, a well-established method suited for assessing sleep in patients' everyday life. To our best knowledge, this is the first study using actigraphy in a domestic context to investigate sleep patterns in a mixed FND cohort. Only two other studies performed actigraphy-based sleep monitoring: Mousa, et al. [34] measured domestic sleep using a wrist-worn accelerometer to compare FS to normative HC and observed increased wakefulness after sleep onset and reduced SE in FS, supporting our results of disrupted sleep in FND. In contrast, Latreille, et al. [33] compared FS patients to epilepsy patients and did not find differences in actigraphy-retrieved sleep patterns. However, this second study was performed during a stay at an epilepsy monitory unit (characterised by regulated day structure and unfamiliar setting) and was restricted to a smaller sample (17 FS, 10 epilepsy), which might account for the diverging results. Our study design did not allow for comparison of actigraphy in FND to a control cohort; instead, we assessed patients' sleep to established criteria for healthy sleep. This revealed SE (in 73%) and TST (in 58%) as disturbed based on thresholds outlining clinically meaningful sleep impairments [30,31].

Besides actigraphy, polysomnography is a popular tool to obtain objective sleep data, providing insight into sleep stages and precise physiological processes. In FND, previous studies using polysomnography have identified alterations in specific sleep parameters in FS compared to epilepsy, including increased SL [33], increased proportion of rapid-eye-movement (REM) sleep [35] or reduced proportion of non-REM sleep stage three [36]. A recent study additionally revealed reduced sleep time, along with increased REM-SL and generally increased SL in FND patients with sleepiness compared to central hypersomnia patients [7]. To date, the available studies on polysomnography in FND remain contradictory and marked by methodological heterogeneity [6]. Actigraphy, as used in our study, does not record sleep stages, but increased SL presents a shared finding in our results and the above-mentioned polysomnography studies [7,33].

4.2. Sleep and comorbid psychiatric symptoms

Given debates on the link between sleep disturbances and psychiatric comorbidities in FND, and their possible aetiological role in poor sleep quality, we explored this association in our data. When accounting for depression and anxiety scores in regression analyses (Table 3), sleep quality impairments seem to be better explained by the inclusion of these psychiatric symptoms than by group alone, hinting towards sleep problems indeed being associated with mood disorder symptoms, except for sleep medication use and numeric SL. A similar link was observed in a previous study that found significantly different sleep patterns in FS compared to epilepsy, but none when comparing only patients with comorbid depression [9]. Furthermore, increased REM sleep proportions were observed in FS patients [35], which resembles sleep patterns in patients with major depression, and may therefore represent a polysomnographic marker corresponding to the comorbid depression in FS. Interestingly, this REM sleep architecture was found under a protocol including stopping of antidepressant and anticonvulsant medication, possibly leading to pronounced effects of depression, whereas a polysomnography study on FS under continuation of antidepressants didn't observe sleep architecture changes [33]. In contrast, another study found that the association of sleep disturbances and functional impairment remained stable even after accounting for anxiety and depression [8].

Current sleep literature postulates a causal influence between sleep quality and psychiatric symptoms to be bidirectional or circular: while reduced sleep quality may impair psychological health, depression and anxiety are known to cause sleep problems [13] – described as a vicious cycle in other disorders [37]. Taken together, manifestations of sleep disturbances may be reinforced by mood disorders in FND patients, whereas disturbed sleep impacts daily life functioning independent of depression or anxiety.

4.3. Inter-assessment discordance in FND

Addressing the debate on a potential subjective rather than objective nature of sleep complaints, we compared PSQI reports to objective actigraphy and found a significant inter-assessment discordance in SL in FND patients, i.e. patients reporting longer duration to fall asleep than measured by actigraphy, but not in TST or SE.

The heterogeneity in objective sleep markers in parallel to recurring subjective impairments led previous authors to conclude that sleep impairments in FND patients may rather represent a functional bias than being rooted in a pathophysiological nature [33]. In support of this, previous studies have indeed found differences in subjective and objective outcome measures [38], specifically when using actigraphy [39] or in sleep studies [7], interpreted as a bias towards over-report of somatic symptoms in FND. However, recent research identified impaired interoceptive skills in FND, resulting in altered integration and perception of bodily signals in line with a Bayesian brain framework [40]. As such, the present formulation of FND proposes that patients may hold overweighted (i.e. overly precise) prior beliefs about the body's state, while interoceptive (i.e. bottom-up sensory) signals are assumed to be down-weighted in precision relative to the priors. As a result, prior beliefs dominate the perceptual inference and eventually shape the patient's subjective experience [41]. Several studies observed altered processing and weighting of interoceptive or emotional signals in FND, supporting the notion of aberrant integration of bottom-up information [42–46]. Translated to sleep, FND patients may have overly precise prior beliefs about poor sleep, which could, in combination with attenuated interoceptive precision on bodily signals (e.g., related to sleep onset), bias the perception and lead to the subjective experience of reduced sleep quality or, as in our study, the subjective experience of prolonged SL. Following this perspective, subjective reports should therefore not be mistaken for inaccurate complaints, but rather as a genuine perception that causes serious burdens on patients' quality of life and could be addressed through therapeutic routes.

From a technical perspective, the accelerometer may underestimate sleep latency due to misclassification of reduced motor activity during pre-sleep wakefulness as the onset of genuine sleep [14]. Compared to PSG, the reference standard in sleep assessments, actigraphy-derived SL has indeed been shown to be less precise (e.g., in patients with depression and insomnia [47]), whereas a meta-analysis from the American Academy of Sleep Medicine (AASM) suggests comparable SL measures between actigraphy and PSG [48]. These methodological considerations call for cautious interpretation of the observed inter-assessment discrepancy and highlight the need for future research to additionally include actigraphy or PSG from controls to disentangle patient-associated effects from general measurement-related divergence.

Taken together, our findings of reduced sleep quality in FND in combination with the potential bilateral influence between mood symptoms and sleep quality and the hypothesised effects of attenuated interoceptive precision in the perception of bodily signals, may lead to experienced poor sleep quality in this patient cohort and a mechanistic understanding of this interplay requires carefully designed studies in the future.

4.4. Strengths

Firstly, the use of self-reports and sensor-based data acquisition allowed to disentangle subjective experience and objective measures in FND patients and age- and sex-matched HC. We emphasise the relevance of both approaches [49], as technology-based measures allow circumventing the 'jumping to conclusion' bias [50], whereas patient-centred questionnaires allow assessing the experienced functional impairment. Moreover, with this approach, we could quantify the discrepancy between patients' perceptions and the objective marker. Secondly, our data was acquired in the patients' homes, and therefore – in contrast to studies performing sleep assessments in synthetic hospital settings - stems from a natural environment with high ecological validity. Thirdly, we compared the FND patients' sleep data to a healthy control cohort (subjective), but also to literature-based thresholds for distinguishing healthy from impaired sleep (subjective and objective), allowing us to exceed relative differences and reveal clinically meaningful impairments. Lastly, in contrast to previously conducted studies on sleep parameters in FND and to the only study with a similar methodological setup [34], we extended our sample to a cohort with mixed functional symptoms instead of FS only and acquired data over a longer timeperiod (14 vs. 7 days) in a higher number of patients (26 vs. 17).

4.5. Limitations

The data acquisition took place over a long timespan, covering varying seasons and periods of the Sars-Cov-19 pandemic under local health protection regulations, which assumingly affected participants' daily schedules, stress levels and sleep behaviours. Although none of the participants had a diagnosis related to sleep disorders, our study did not assess data to exclude biological origins of sleep disturbances, such as anaemia, thyroid disorder or other comorbidities in functional disorders (e.g., sleep-disordered breathing or periodic limb movement) [36,51], and we cannot exclude pharmacological effects caused by medication. Furthermore, we did not perform psychiatric interviews to obtain validated psychiatric diagnoses, limiting our conclusions on the effects of mood disorders to self-reports. While actigraphy is a valid method to assess sleep patterns, it doesn't allow insights into detailed sleep architecture and precise read-outs, as in polysomnography. Crucially, actigraphy was collected exclusively from FND patients, preventing between-group analyses. In consequence, it remains unclear whether the findings on objective sleep patterns or inter-assessment concordance are FND-specific, which requires future studies to include comparisons to a healthy control group, over multiple nights, adhering to AASM recommendations [52]. Lastly, due to the limited sample, stratification based

on symptom type (e.g., FS vs. motor symptoms) were not possible and the presented results are marked with small to medium effect sizes, requiring replication in larger studies to strengthen the understanding of the complex interplay of sleep, FND, comorbidities, and perceptual mismatches.

4.6. Implications

The presented data indicates severe sleep quality complaints in FND patients, not only in comparison to HC (subjective) but also to thresholds delineating healthy sleep (subjective and objective), which should encourage researchers to engage in further investigations. Next steps should include a study using domestically assessed objective outcomes (actigraphy, polysomnography) and detailed subjective evaluations in mixed FND patients, including a broad range of sleep-affecting variables (e.g., medication, comorbidities, daily symptom burden, sleep hygiene, daytime structure) to shed light on detailed sleep architecture and origins of poor sleep. Clinical care may benefit from including sleep quality in clinical anamnesis and discussions on treatment choices, which may target sleep management (e.g., cognitive behavioural therapy for insomnia, CBI-I; promising outcomes in various cohorts [53–55]), or psychoeducation and mindfulness-based programs to improve the perception of bodily states and align subjective perception with objective sleep quality.

5. Conclusion

In summary, our findings revealed major sleep complaints in patients with FND that might affect their quality of life and potentially derogate the chances of recovery. Furthermore, we observed a disparity between patients' self-reports and actigraphy, potentially pointing towards a misconception of one's own sleep quality that could serve as a treatment target. We advocate for increased focus on sleep disturbances in clinical practice, while future studies should aim at further disentangling the role of sleep in this patient cohort.

CRediT authorship contribution statement

Janine Bühler: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Nathalie Schwab:** Writing – review & editing, Writing – original draft, Formal analysis. **Fabian Messmer:** Writing – review & editing, Validation, Investigation, Formal analysis, Data curation, Conceptualization. **Samantha Weber:** Writing – review & editing, Project administration, Methodology, Investigation, Conceptualization. **Julia van der Meer:** Writing – review & editing, Resources, Methodology, Conceptualization. **Selma Aybek:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

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Declaration of competing interest

none of the authors declare any relevant competing interests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jpsychores.2026.112634.

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